PART-A

1.

i) OSI Model

- It has 7 layers, it is a theoretical model
- It has strict layer seperation.
- Example protocols: FTP, HTTP, TCP, IP.
- Developed by ISO.

TCP / IP Model

- It has 4 layeprs,
- It is practical Model
- Layer often overlap
- Protocols are directly included and developed by DARPA.

ii) Layer mapping :-

OSI :

These OSI use Application, Presentation, Session, and for transport, network and Data - link and physical.

TCP / IP :

This is an application, ~~T~~ Transport and Internet and Network Access.

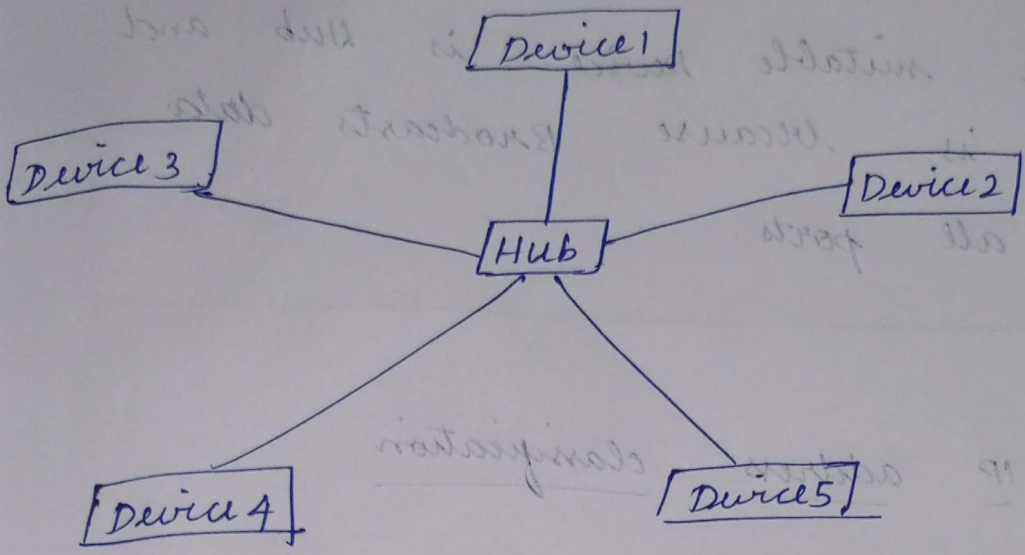
2.

- ATM network within one city it uses MAN (Metropolitan Area Network)
- Google's Worldwide backbone it uses WAN (Wide Area Network)
- School computer ~~tap~~ lab it uses LAN (Local Area Network)

3.

It uses star topology. Because it is easy to add / remove computers and it is High performance and Fault isolation possible and Installation cost is moderate

Diagram: Star Topology



4.

i) To Regenerate weak signals and the suitable device is repeater it is because it Amplifies signals strength.

ii) connect ~~to~~ two LAN segments the Bridge is the suitable device. and it is because connects two LAN's data link layer.

iii) Reduce collisions and improve Performance and the suitable device is, switch and it because it creates a seperate collision domains.

iv) Broadcast data to all computers the suitable Device is Hub and it is because Brodcasts data to all ports.

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5)

IP address classification

IP Address	class	Range
192. 168. 1.5	class C	192 - 223
130·25.10	class B	128 - 191
10.0.0.7	class A	1 - 129.

Explanation of classes:-

class	Range	~~Use~~
A	1.0.0.0 - 126.255.255.255	
B	128.0.0.0 - 191.255.255.255	
C	192.0.0.0 - 223.255.255.255	